Types of JOIN in SQL

Lab 4 -

Inner Join

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;
CREATE TABLE Users ( user_id INT PRIMARY KEY, name VARCHAR(50), email
VARCHAR(100) );
CREATE TABLE Orders ( order_id INT PRIMARY KEY, user_id INT, product_name
VARCHAR(100), amount DECIMAL(10, 2), FOREIGN KEY (user_id) REFERENCES
Users(user_id) );
INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sanjana', 'sanjana@gmail.com'),
(2, 'Sahasra', 'sahasra@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');
INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES
(101, 1, 'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);
SELECT
Users.user_id,
Users.name,
Users.email,
Orders.order_id,
Orders.product_name,
Orders.amount FROM Users INNER JOIN Orders ON Users.user_id = Orders.user_id;
```

Output:

Output:

```
+---------+---------+-------------------+----------+--------------+--------+
| user_id | name    | email             | order_id | product_name | amount |
+---------+---------+-------------------+----------+--------------+--------+
|       1 | Sanjana | sanjana@gmail.com |      101 | Smartphone   | 699.99 |
|       1 | Sanjana | sanjana@gmail.com |      102 | Headphones   | 150.00 |
|       2 | Sahasra | sahasra@gmail.com |      103 | Smartwatch   | 249.50 |
+---------+---------+-------------------+----------+--------------+--------+
```

Left join-

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;

DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2),
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sahasra', 'sahasra@gmail.com'),
(2, 'Ananya', 'ananya@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES
(101, 1, 'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);

SELECT
    Users.user_id,
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
LEFT JOIN Orders ON Users.user_id = Orders.user_id;
```

Output:

Output:

```
+---------+---------+-------------------+----------+--------------+--------+
| user_id | name    | email             | order_id | product_name | amount |
+---------+---------+-------------------+----------+--------------+--------+
|       1 | Sahasra | sahasra@gmail.com |      101 | Smartphone   | 699.99 |
|       1 | Sahasra | sahasra@gmail.com |      102 | Headphones   | 150.00 |
|       2 | Ananya  | ananya@gmail.com  |      103 | Smartwatch   | 249.50 |
|       3 | Ninada  | ninada@gmail.com  |     NULL | NULL         |   NULL |
+---------+---------+-------------------+----------+--------------+--------+
```

Outer Left Join (UNION of Left and Right Joins) -

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;

DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2)
);

INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sahasra', 'sahasra@gmail.com'),
(2, 'Ananya', 'ananya@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES
(101, 1, 'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50),
```

(104, 4, 'Tablet', 300.00);

```
SELECT
    Users.user_id,
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
LEFT JOIN Orders ON Users.user_id = Orders.user_id
UNION
SELECT
    Users.user_id,
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
RIGHT JOIN Orders ON Users.user_id = Orders.user_id;
```

Output:

Output:

user_id	name	email	order_id	product_name	amount
1	Sahasra	sahasra@gmail.com	102	Headphones	150.00
1	Sahasra	sahasra@gmail.com	101	Smartphone	699.99
2	Ananya	ananya@gmail.com	103	Smartwatch	249.50
3	Ninada	ninada@gmail.com	NULL	NULL	NULL
NULL	NULL	NULL	104	Tablet	300.00

Cross join -

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;

DROP TABLE IF EXISTS Orders;
```

```
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2)
);

INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sahasra', 'sahasra@gmail.com'),
(2, 'Ananya', 'ananya@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES
(101, 1, 'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);

SELECT
    Users.user_id,
    Users.name,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
CROSS JOIN Orders;
```

Output:

Output:

user_id	name	order_id	product_name	amount
3	Ninada	101	Smartphone	699.99
2	Ananya	101	Smartphone	699.99
1	Sahasra	101	Smartphone	699.99
3	Ninada	102	Headphones	150.00
2	Ananya	102	Headphones	150.00
1	Sahasra	102	Headphones	150.00
3	Ninada	103	Smartwatch	249.50
2	Ananya	103	Smartwatch	249.50
1	Sahasra	103	Smartwatch	249.50

Subquery (Find teachers whose salary is greater than the average salary) -

```
CREATE DATABASE IF NOT EXISTS school_db;
USE school_db;

DROP TABLE IF EXISTS Teachers;

CREATE TABLE Teachers (
    teacher_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100),
    subject VARCHAR(50),
    salary DECIMAL(10, 2)
);

INSERT INTO Teachers (teacher_id, name, email, subject, salary) VALUES
(1, 'Sahasra', 'sahasra@school.edu', 'Mathematics', 65000.00),
(2, 'Ananya', 'ananya@school.edu', 'Science', 72000.00),
(3, 'Ninada', 'ninada@school.edu', 'English', 58000.00);

SELECT
    teacher_id,
    name,
    subject,
    salary
FROM Teachers
WHERE salary > (
    SELECT AVG(salary)
    FROM Teachers);
```

Output:

Output:

```
+------------+--------+---------+----------+
| teacher_id | name   | subject | salary   |
+------------+--------+---------+----------+
|          2 | Ananya | Science | 72000.00 |
+------------+--------+---------+----------+
```